ModelScope

Project Report — Team 011-03

CSCI 3308 — University of Colorado Boulder

Who

Team Number	011-03
Team Members	Charlie Coop, Emily McIntyre, Wallis McGuire, Heloisa D'Incao, and Justin Sorrells
Course	CSCI 3308
Institution	University of Colorado Boulder

Project Description

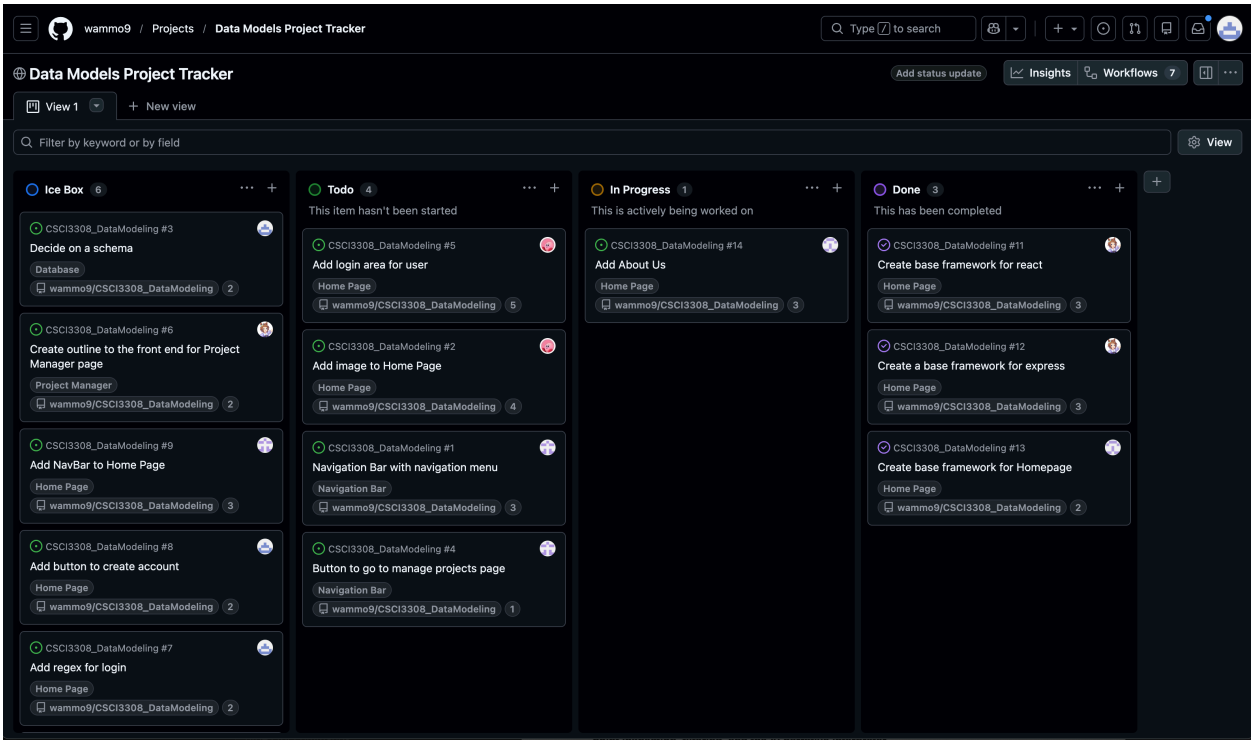
ModelScope is a full-stack web application designed for uploading, analyzing, and visualizing CSV datasets through Principal Component Analysis (PCA). The platform enables users to upload CSV files and receive immediate automated data profiling — detecting numeric and categorical columns, flagging missing values, identifying constant or mixed-type columns, and assessing dataset readiness for PCA.

Users configure their analysis by selecting features and setting preprocessing options such as missing value imputation (drop, mean, or median), outlier removal (z-score or IQR-based), and feature standardization. The server computes PCA using Singular Value Decomposition, reducing high-dimensional data into 2 or 3 principal components. Results are displayed as interactive 2D or 3D scatter plots, a scree plot showing variance explained per component, and a feature loadings chart identifying which original variables drive each component.

Additional features include K-means clustering on PCA output, point inspection by clicking individual data points, advanced filtering by label, numeric range, and cluster assignment, correlation heatmaps, distribution histograms with box plots, dataset versioning, saved configuration presets, and AI-powered narrative interpretation of PCA results using the Anthropic Claude API. The application is built with React and Plotly.js on the frontend, Node.js and Express on the backend, and PostgreSQL for persistent storage, containerized with Docker and deployed on Render.

Project Tracker — GitHub Project Board

https://github.com/users/wammo9/projects/1



Video Demo

https://github.com/wammo9/CSCI3308_DataModeling/blob/main/projectDemoSmall.mov

The video demonstrates the full ModelScope workflow: user registration and login, CSV upload and automatic data profiling, PCA configuration and execution, interactive 2D/3D scatter plot visualization, scree plot and feature loadings inspection, K-means clustering, point inspection, filtering, and the AI narrative interpreter.

VCS — Version Control Repository

https://github.com/wammo9/CSCI3308_DataModeling

The repository contains:

- Source Code (client/ and server/ directories)
- Test Cases (server/test/ and projectSourceCode/test/)
- Video Demo (projectDemoSmall.mov)
- README.md with setup and deployment instructions
- Project Documentation (milestones/ and milestoneSubmissions/)
- GitHub Project Board (linked above)

Contributions

Emily — Frontend Development

Emily led all frontend development for ModelScope. She built the React single-page application using Vite, implementing the Login, Register, Projects, and Visualize pages. Emily designed and coded the full CSS design token system including typography, color palette, spacing, and component styles. She integrated Plotly.js to render the correlation heatmap, histogram, box plot, and scatter plot charts. Emily was responsible for the overall look and feel of the product.

Wallis — Project Management

Wallis served as project manager for Team 011-03. She coordinated milestone submissions, maintained the GitHub project board, and ensured the team stayed on schedule throughout the semester. Wallis prepared and submitted milestone documentation including wireframes, the use case diagram, and the user acceptance test plan. She facilitated communication across frontend and backend team members and organized project artifacts in the repository.

Charlie — Backend Development

Charlie worked on backend development using Node.js and Express. He contributed to API route implementation, the authentication system using JWT and bcrypt, and the CSV parsing pipeline using Multer and csv-parse. Charlie participated in writing the user acceptance test plan and conducted UAT testing. He collaborated closely with Helo and Justin on backend architecture decisions and helped debug deployment issues on Render.

Helo — Backend Development

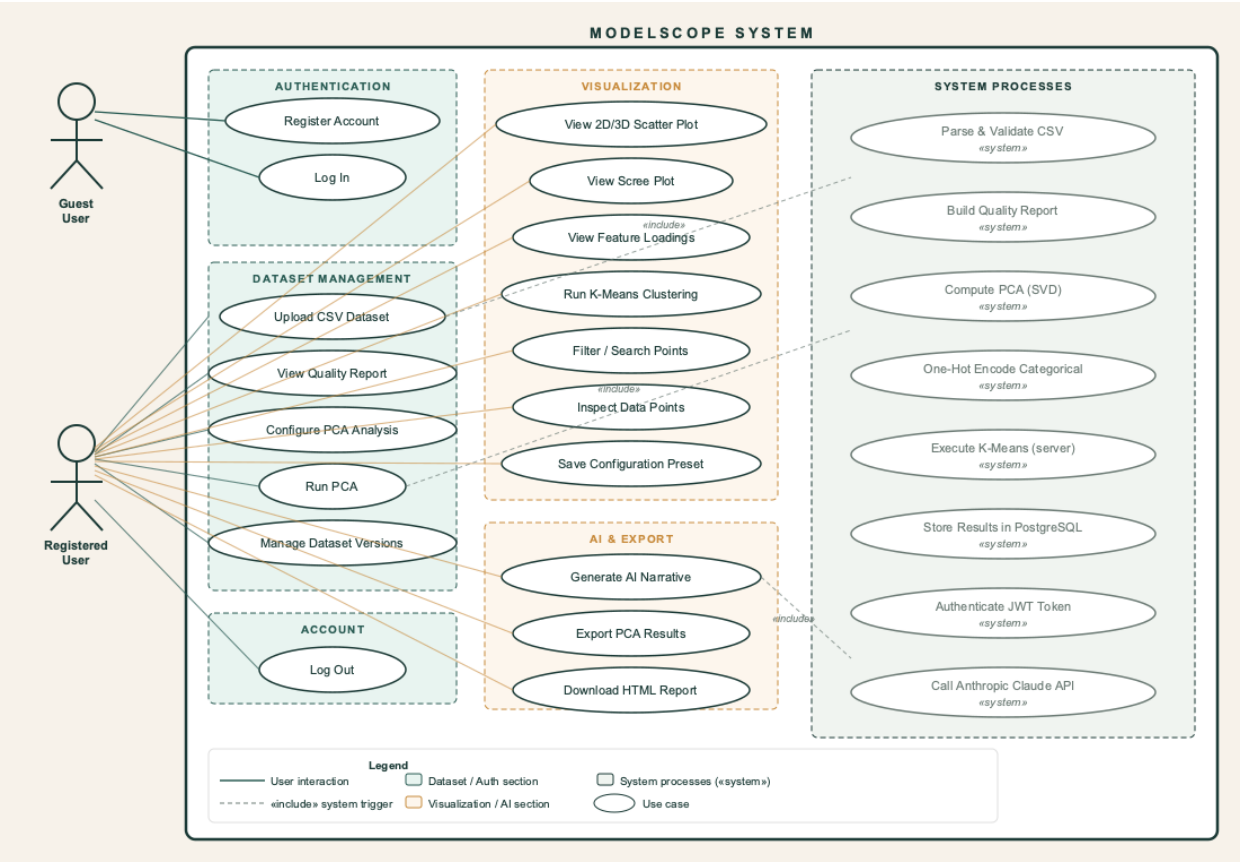
Helo contributed to backend development with a focus on the data processing pipeline. He worked on the quality report system that automatically profiles uploaded CSV datasets, detecting missing values, constant columns, and mixed-type columns at upload time. Helo also contributed to the preprocessing pipeline including missing value imputation strategies, outlier removal using z-score and IQR methods, and the preprocessing diff summary shown to users after each PCA run.

Justin — Backend Development

Justin worked on backend development with a focus on the PCA computation engine and database architecture. He implemented the PCA pipeline using SVD, K-means clustering, Pearson correlation matrix computation, and the dataset versioning system. Justin configured Docker Compose for local development, implemented the Render deployment with static file serving, fixed the database initialization issue for cloud deployment, and maintained the 37-test automated test suite.

Use Case Diagram

The diagram below shows all actors and use cases in the ModelScope system. Guest Users can register and log in. Registered Users have access to all data management, visualization, AI, and export features. System processes are triggered automatically in response to user actions.



Wireframes

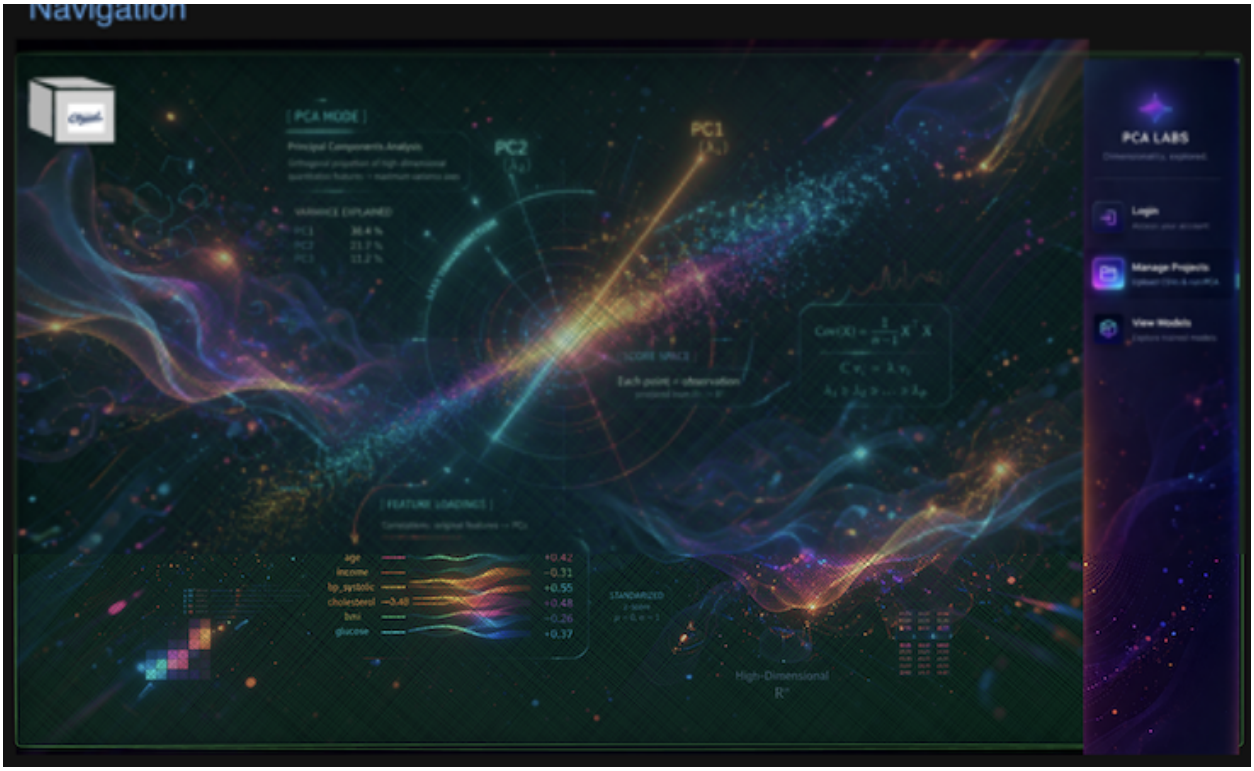
The following wireframes were created during the project proposal and design phase. All wireframe images are stored in the milestoneSubmissions/ directory of the repository.

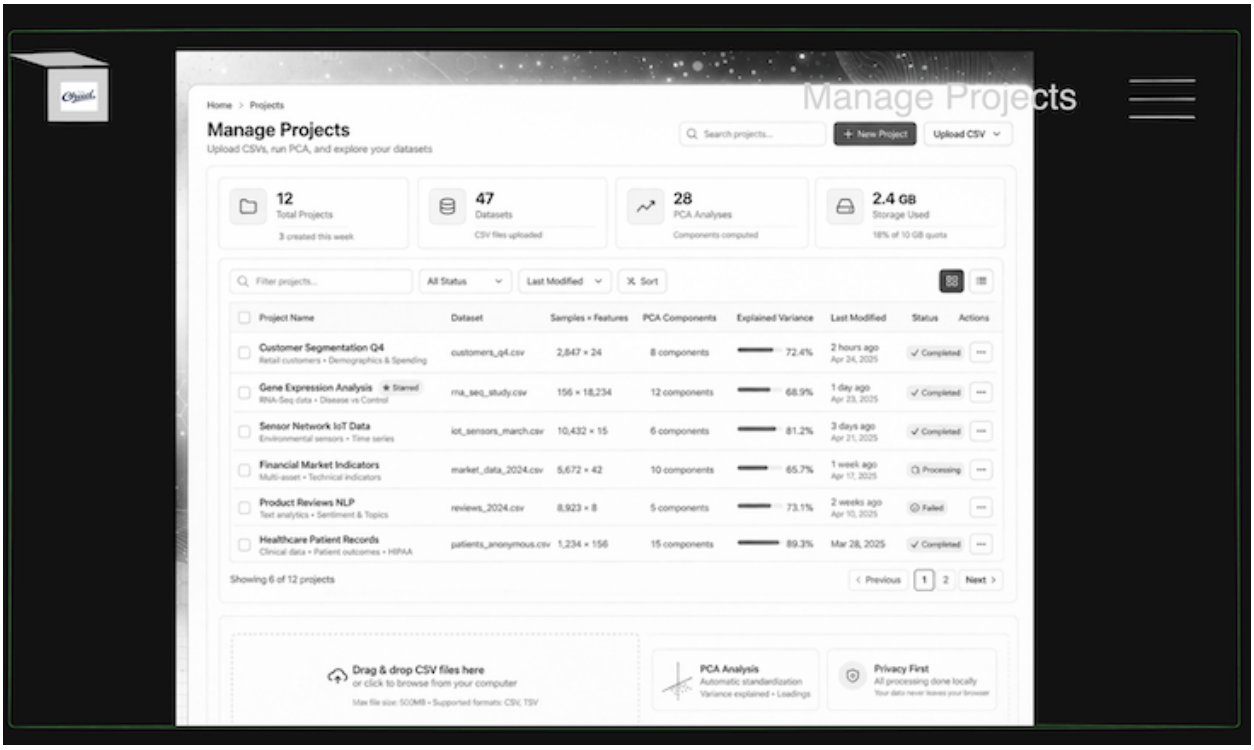
https://github.com/wammo9/CSCI3308_DataModeling/tree/main/milestoneSubmissions

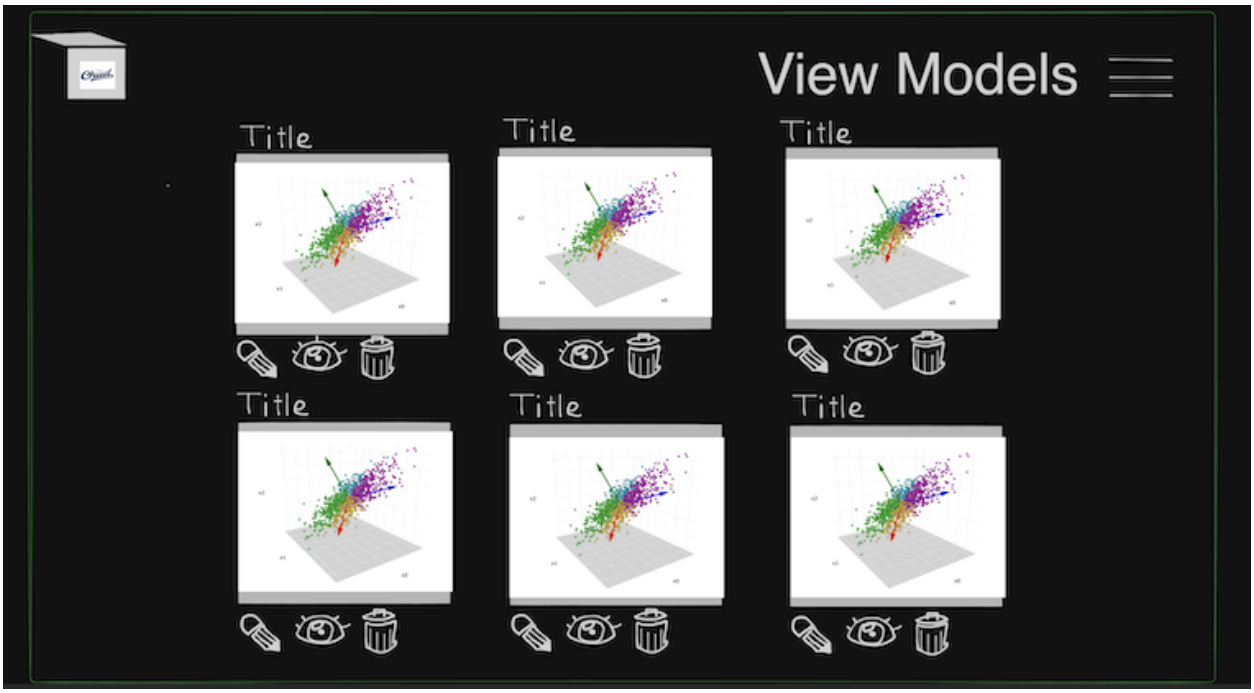
Wireframes included:

- Home Page (home_page.png)

- Manage Datasets Page (manage_pages.png)
- View Models Page (view_models_page.png)
- Visualization Page (visualization_page.png)
- General Wireframe Overview (wireframe.png)
- UML Data Model Diagram (uml_dataModeling.png)






View Models
Title
Title
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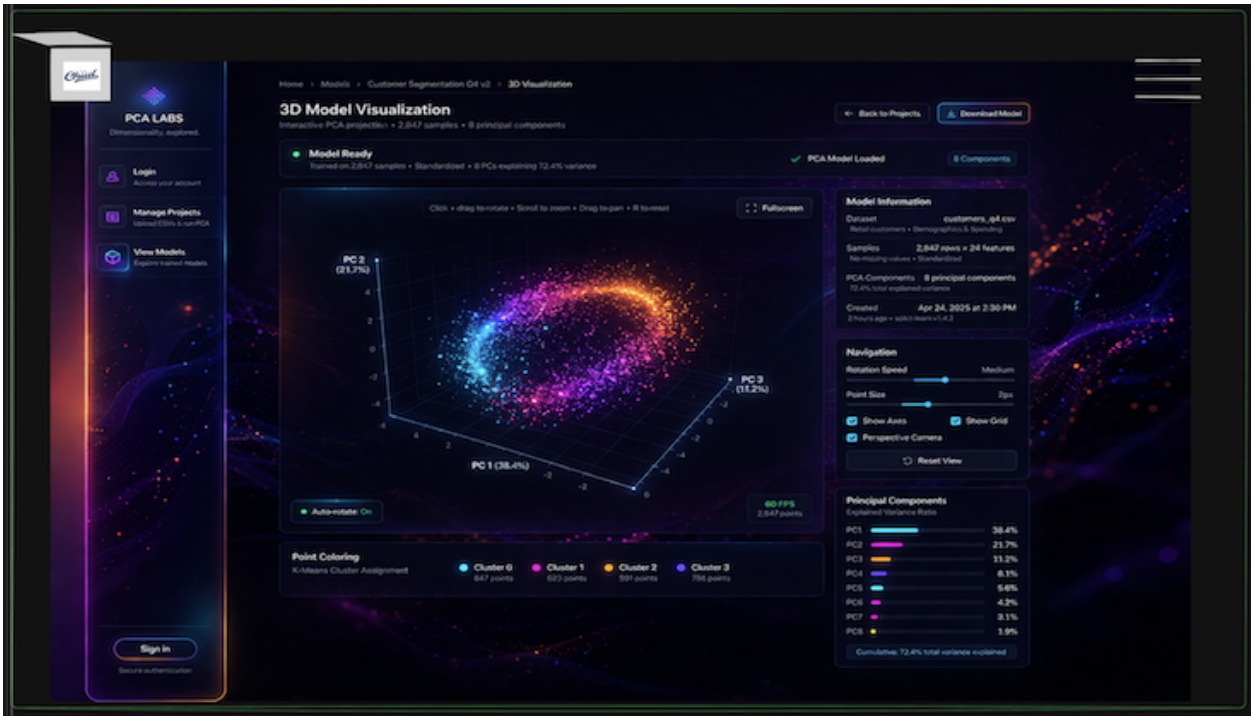
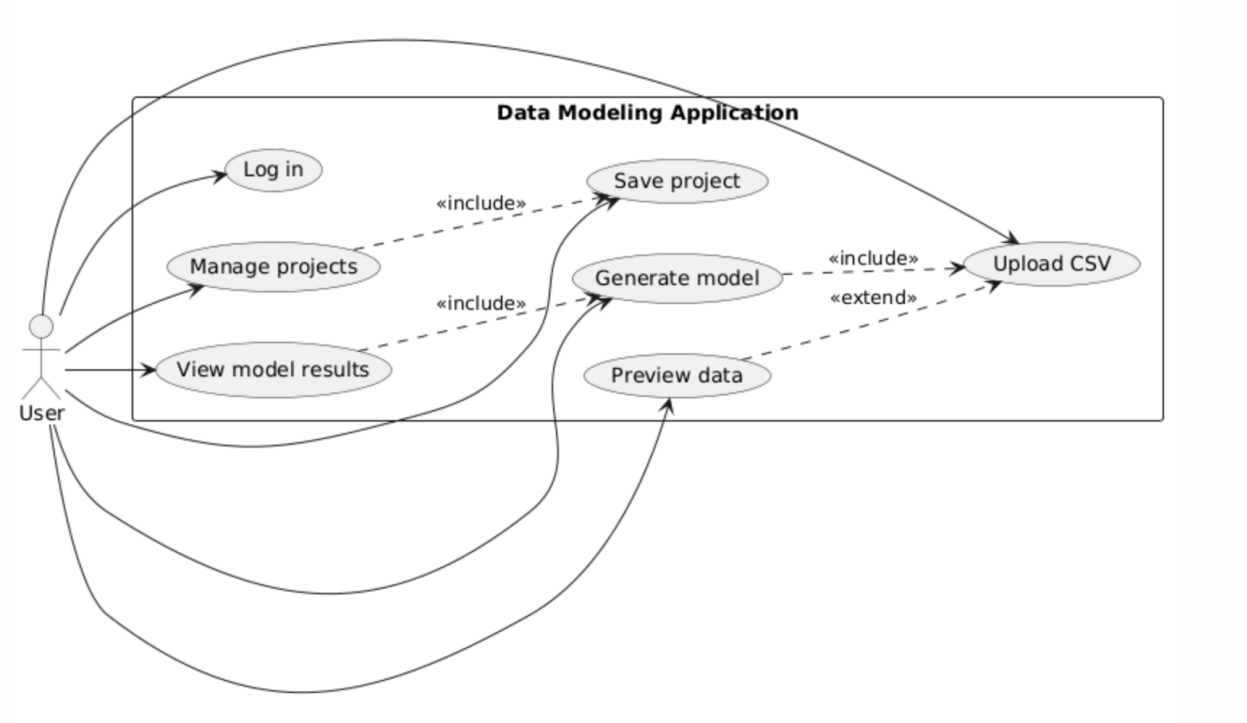

PCA LABS
Login
Manage Projects
View Models
Sign in
3D Model Visualization
Back to Projects
Download Model
Model Ready
PCA Model Loaded
Fullscreen
PC 2 (21.7%)
PC 3 (11.2%)
PC 1 (38.4%)
Model Information
Navigation
Reset View
Principal Components
Point Coloring
Cluster 0
Cluster 1
Cluster 2
Cluster 3



Test Results

User Acceptance Testing was conducted on April 8, 2026 by Charlie and Wallis McGuire against the local development environment (frontend: localhost:5173, backend: localhost:5001).

Test ID	Objective	Result	Resolution	Post-Resolution
UAT-01	API Health Check — GET /api/health returns 200	FAIL — missing cors dependency	Run docker compose up --build	**PASS**
UAT-02	Feature List API — GET /api/features returns feature array	FAIL — blocked by UAT-01 failure	Same as UAT-01	**PASS**
UAT-03	Home Page — frontend renders login page with status 200	FAIL — Vite not found	npm install --include=dev or use Docker	**PASS**

Automated Test Suite Results

- Server tests (Mocha + Supertest): 32 passing
- Lab requirement tests (projectSourceCode/, Mocha + Chai-http): 5 passing
- Total: 37 tests passing, 0 failing

Deployment

Environment	Render.com
Live URL	https://csci3308-datamodeling.onrender.com/
Build Command	cd client && npm install --include=dev && npm run build && cd ../server && npm install
Start Command	node server/src/index.js
Database	Render managed PostgreSQL service

Environment Variables

- DATABASE_URL — Render internal PostgreSQL connection string
- JWT_SECRET — Secret key for signing authentication tokens
- NODE_ENV=production
- ANTHROPIC_API_KEY — Required for AI narrative feature

To Run Locally

docker compose down -v && docker compose up --build

Then visit http://127.0.0.1:5173 in your browser.